

Docking Universal - Docking Study Report

Version 0.4.0 - Research preview

1. Configured docking protocol

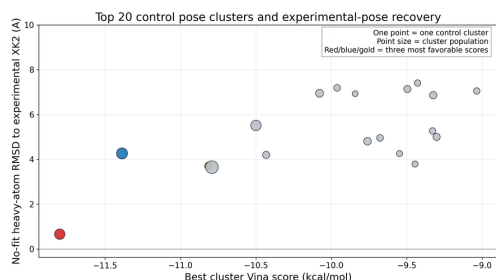
This summary identifies the approved target-matched protocol and the control result that authorized its reuse.

Control-validation summary	Result
Protocol status	PASS
Control completed	2026-08-18T21:06:10.283097+00:00
Control ligand	XK2
Control ligand RMSD (best sampled)	0.5788 A
Acceptance threshold	2.0 A
Locked docking box / pocket	1HVR_pocket1.conf

2. Experimental control redocking results: XK2

This retrospective control tests whether the protocol reproducibly recovers the experimental pose of XK2. PASS requires sampling, ranking, and independent-seed criteria to pass. It supports use of the selected protocol for this target, but does not establish affinity or prospective pose accuracy.

A



B

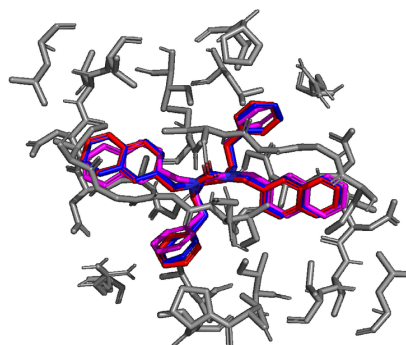


Figure 1. Retrospective control performance and pose recovery. (A) Control-cluster Vina score versus symmetry-aware, no-fit heavy-atom RMSD to experimental XK2 in the receptor coordinate frame. (B) Experimental ligand (magenta), lowest-energy pose (red), and lowest-RMSD pose (blue) superimposed in that frame; receptor residues within 5 Å of the displayed ligands are gray.

3. New-ligand docking results

Target: 1HVR | Ligand: Rilpivirine

Docking used the approved target-matched protocol shown above. More favorable Vina scores are more negative; RMSD and cluster population are separate measures.

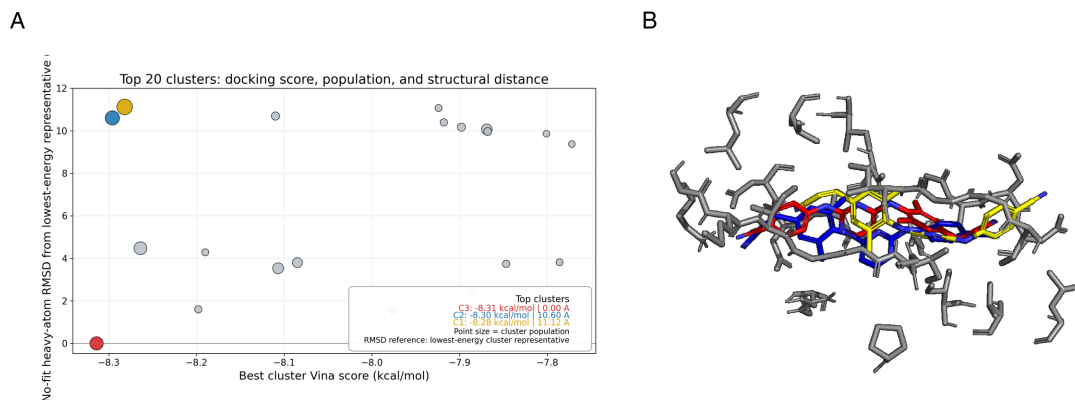


Figure 2. Docking pose-cluster analysis for Rilpivirine. (A) Docking score versus symmetry-aware, no-fit heavy-atom RMSD from the lowest-energy cluster representative in the receptor coordinate frame; point size denotes cluster population. (B) Representative structures from the highlighted clusters, using matching cluster colors.

Top-ranked 3D cluster snapshots

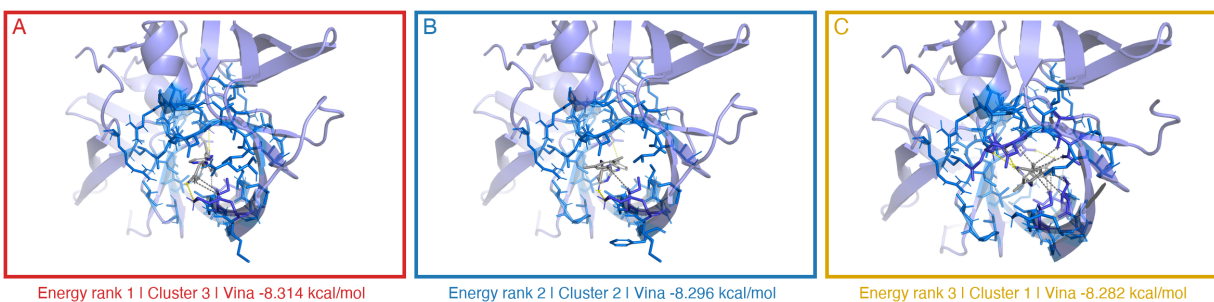


Figure 3. Three-dimensional interaction snapshots for Rilpivirine. Shown are 3 energy-ranked distinct cluster representatives, ordered by Vina score. Red, blue, and gold match the highlighted clusters in Figure 2. These views support structural inspection; docking score rank does not establish pose correctness.

Top-ranked pose interaction diagrams

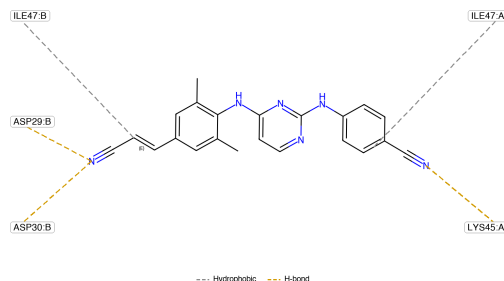


Figure 4. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 1, cluster 3 (Vina score -8.314 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

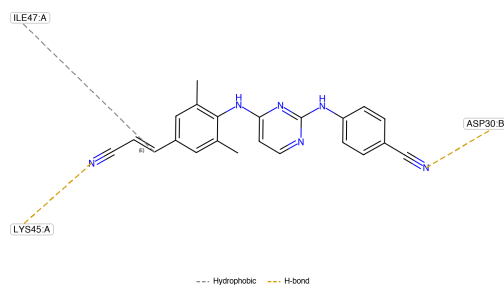


Figure 5. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 2, cluster 2 (Vina score -8.296 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

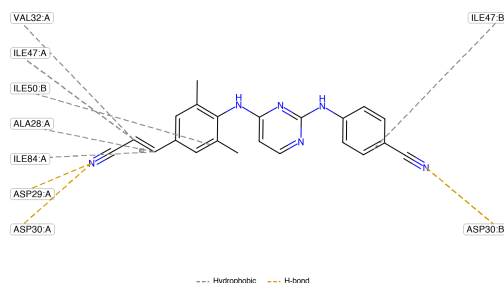


Figure 6. SDF-aware PLIP interaction diagram for Rilpivirine, energy rank 3, cluster 1 (Vina score -8.282 kcal/mol). Ligand bond orders, aromaticity, formal charges, and stereochemistry come from the retained pose SDF; interaction calls come from the retained PLIP report.xml. The PLIP XML/text files remain the authoritative interaction records.

Ranked docking clusters

Rank	Cluster	Best score	Median score	Poses	Seeds
1	Cluster 3	-8.314	-8.232	30	5
2	Cluster 2	-8.296	-8.181	35	5
3	Cluster 1	-8.282	-8.188	44	5
4	Cluster 4	-8.264	-8.016	27	5
5	Cluster 20	-8.198	-8.196	3	3
6	Cluster 15	-8.19	-8.153	2	2
7	Cluster 18	-8.11	-7.743	6	4
8	Cluster 13	-8.107	-7.961	17	5
9	Cluster 14	-8.085	-7.895	13	5
10	Cluster 16	-7.977	-7.881	2	1
11	Cluster 5	-7.924	-7.874	2	2
12	Cluster 8	-7.918	-7.842	3	3

Rank	Cluster	Best score	Median score	Poses	Seeds
13	Cluster 17	-7.898	-7.832	6	3
14	Cluster 7	-7.886	-7.808	4	4
15	Cluster 6	-7.869	-7.76	16	4
16	Cluster 9	-7.868	-7.777	5	4
17	Cluster 19	-7.847	-7.692	3	1
18	Cluster 23	-7.801	-7.801	1	1
19	Cluster 11	-7.786	-7.669	2	2
20	Cluster 10	-7.772	-7.772	1	1

Interpretation and limitations

Docking scores are ranking estimates, not measured binding free energies. Rigid-receptor docking does not model induced fit. Cluster population and seed support describe computational convergence, not biological correctness. Protonation, tautomer, receptor preparation, and box choices can affect results. Experimental validation remains necessary.

4. Reproducibility, software, and references

Docking scores and poses were produced by AutoDock Vina. PLIP supplied rule-based protein-ligand interaction calls; the retained PLIP XML is the authoritative interaction record. RDKit supplied the retained molecular graph handling and symmetry-aware, no-fit heavy-atom RMSD matrix used by Butina clustering. The clustering cutoff was 2.0 Å. If only one cluster is present, its lowest-energy member is reported as the sole representative. PyMOL produced the 3D molecular panels; ReportLab assembled this PDF.

Control-to-new-run software check

Checked: no software version differences detected.

Software	Control	New run	Result
Docking Universal	0.4.0	0.4.0	SAME
Python	3.9.23	3.9.23	SAME
RDKit	2023.9.6	2023.9.6	SAME
MolScrub	0.2.2	0.2.2	SAME
Meeko	0.6.1	0.6.1	SAME
AutoDock Vina	AutoDock Vina v1.2.7	AutoDock Vina v1.2.7	SAME

Additional software used for analysis and reporting

Result element	Software	Version used
Molecular conversion/PLIP backend	Open Babel	3.1.0
Interaction calls	PLIP	2.3.1
3D rendering	PyMOL	3.0.0
Plots	Matplotlib	3.9.4
PDF generation	ReportLab	5.0.0

Scientific and software references

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2. Le Guilloux V, Schmidtke P, Tuffery P. Fpocket: an open source platform for ligand pocket detection. BMC Bioinformatics. 2009;10:168. <https://doi.org/10.1186/1471-2105-10-168>
3. Santos-Martins D, He Y, Eberhardt J, et al. Meeko: molecule parameterization and software interoperability for docking and beyond. J Chem Inf Model. 2025;65:13045-13050. <https://doi.org/10.1021/acs.jcim.5c02271>
4. Salentin S, Schreiber S, Haupt VJ, Adasme MF, Schroeder M. PLIP: fully automated protein-ligand interaction profiler. Nucleic Acids Res. 2015;43:W443-W447. <https://doi.org/10.1093/nar/gkv315>
5. Butina D. Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. J Chem Inf Comput Sci. 1999;39:747-750. <https://doi.org/10.1021/ci9803381>
6. O'Boyle NM, Banck M, James CA, Morley C, Vandermeersch T, Hutchison GR. Open Babel: An open chemical toolbox. J Cheminform. 2011;3:33. <https://doi.org/10.1186/1758-2946-3-33>
7. RDKit: Open-source cheminformatics software. <https://www.rdkit.org/>
8. The PyMOL Molecular Graphics System, Version 3.0.0, Schrodinger, LLC. <https://www.pymol.org/support.html>